College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 1 Year: 2024

Subject: 031021121 Electric Circuits I Section: 05-06

Date: 8 October 2024 Time: 12:00-15:00

Name:______________________________ID:____________Field of Study:___________________

Directions: The test is designed to measure your comprehension. The test is divided into 3 sections. There will be 9 pages (including this page) and they are worth 50 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University

This exam paper is part of a set used to assess student abilities as follows:		Part 1	Part 2	Part 3
CLO 1	Explain the Current Source, Source Conversions, Current Sources in Parallel, Current Sources in Series, Capacitors and Inductors	√		
CLO 2	Identify the types of resistor DC networks.	√	√	
CLO 3	Analyze the resistor DC networks using Mesh Analysis, Nodal Analysis, Branch Current Analysis, Superposition's Theorem, Thevinin's Theorem, Norton's Theorem and Maximum power transfer theorem		√	√
CLO 4	Design the resistor DC networks using Mesh Analysis, Nodal Analysis, Branch Current Analysis, Superposition's Theorem, Thevinin's Theorem, Norton's Theorem and Maximum power transfer theorem		√	√

Part 1 (10 points) True/False

Instruction: Mark "True" or "False" for the following questions in **THE PROVIDED ANSWER SHEET (SEE LAST PAGE)**.

1. When using the superposition theorem, an ideal voltage must be replaced by an open circuit.
2. Thevenin's theorem permits the reduction of any two-terminal linear circuit to a single voltage source and series resistance.
3. For any physical network, the value of E_{Th} can be determined experimentally by measuring the open-circuit voltage across the load terminals.
4. The total capacitance for inductors in series and parallel can be found the same way as resistors in series and parallel.
5. All practical sources have some internal resistance.
6. For loads connected directly to a dc-voltage supply, maximum power will be delivered to the load when the Thévenin resistance is equal to the internal resistance of the source.
7. Source conversions are equivalent at their internal terminals.
8. R_N is different from R_{Th}.
9. An ideal inductor looks like a short circuit to dc current.
10. Short circuit between terminals is equivalent to zero voltage.

Part 2 (20 points) Multiple choices

Instruction: Mark the correct answer for the following questions in **THE PROVIDED ANSWER SHEET**.

1. Find voltage at *a*-*b* terminals.

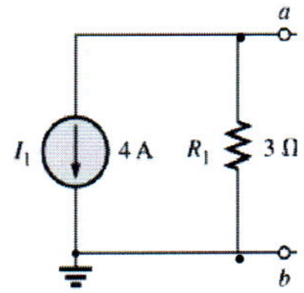


Figure 1.

(A) 12 V and 3 **Ω** (B) 4 V and 3 **Ω** (C) 12 V and 4/3 **Ω** (D) 4 V and 4/3 **Ω**

2. From Figure 1. Convert the current source to a voltage source.

   (A) 12 V and 3 Ω (B) 4 V and 3 Ω (C) 12 V and 4/3 Ω (D) 4 V and 4/3 Ω

3. From Figure 2. Which of the following terms describes the voltage across the 7 Ω resistor when using the mesh analysis?

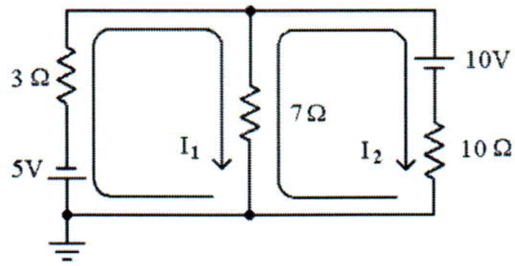


Figure 2.

   (A) $(7\ \Omega)\ I_1$ (B) $(7\ \Omega)\ I_2$

   (C) $(7\ \Omega)(I_1 - I_2)$ (D) $(7\ \Omega)(I_1 + I_2)$

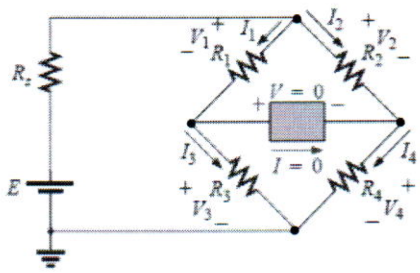


Figure 3.

4. From Figure 3, what is the condition of balance bridge?

   (A) $R_2 / R_1 = R_3 / R_4$ (B) $R_1 R_2 = R_3 R_4$ (C) $R_1 / R_3 = R_2 / R_4$ (D) $R_1 R_3 = R_2 R_4$

5. What the following is correct when all three resistors in Y (star) network are equal?

   (A) $R_\Delta = 3R_Y$ (B) $R_\Delta = 3/R_Y$ (C) $R_\Delta = R_Y$ (D) $R_\Delta = 1/R_Y$

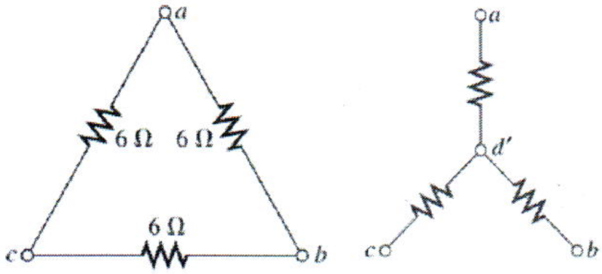
6. From Figure 3. If nodal analysis were used to solve for unknown voltages in this circuit, how many nodes would be needed (including the reference node)?

   (A) 1 (B) 2 (C) 3 (D) 4

7. E_{th} is the ____________ voltage at the two-terminal network.

   (A) short circuit (B) load (C) fully loaded (D) open circuit

8. The most obvious advantage of the superposition theorem is that ____________________ advanced mathematical techniques.



(A) all in 1 Ω (B) all in 2 Ω (C) all in 3 Ω (D) all in 6 Ω

16. Norton's theorem states that the Norton current is equal to_________.

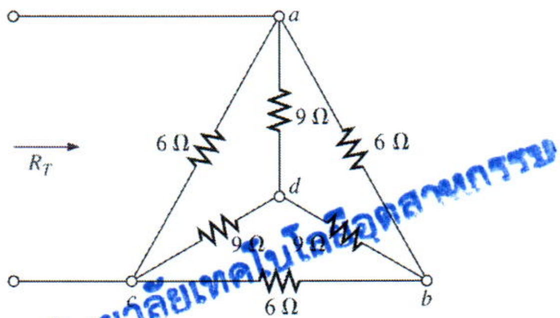
(A) open circuit voltage at the network terminals.

(B) short circuit voltage at the network terminals.

(C) open circuit current at the network terminals.

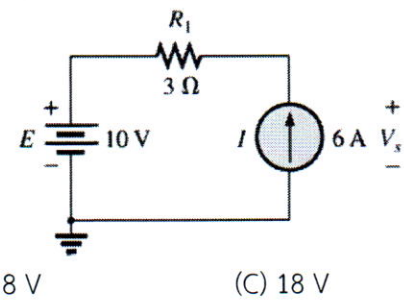
(D) short circuit current at the network terminals.

17. Find the total resistance RT of the figure below.



(A) 3.27 Ω (B) 6.7 Ω (C) 4 Ω (D) 6 Ω

18. Find the value of voltage V_s from the figure below.



(A) 0 V (B) 8 V (C) 18 V (D) 28 V

19. Given linear system of equation below, what is the current I_1 and I_3?

$$15-4I_1+10I_3-20=0$$

$$20-10I_3-5(I_1+I_3)+40=0$$

(A) 4.77A, 2.41A (B) 4.0A, 2.0A (C) 5.77A, 3.41A (D) 6.7A, 2.0A

(A) it makes it harder to use

(B) it makes it easier to use

(C) it does not require

(D) it requires

9. I_N is the ____________ current at the two-terminal network.

(A) short circuit

(B) load

(C) fully loaded

(D) open circuit

10. For loads connected directly to a dc-voltage supply, maximum power will be delivered to the load when the _________________ is equal to the internal resistance of the source.

(A) total resistance

(B) Norton resistance

(C) load resistance

(D) Thévenin resistance

11. What is the value of I_2 ?

$$I_2 = \frac{\begin{vmatrix} 6 & 2 \\ 4 & 6 \end{vmatrix}}{14}$$

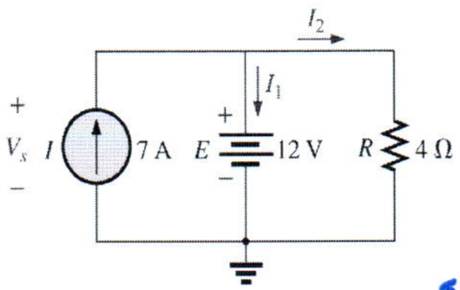
(A) 0 A

(B) 0.5 A

(C) 1 A

(D) 2 A

12. Find the voltage Vs and current I2 for the network in figure below.



(A) 12 V and 3 A

(B) 3 V and 12 A

(C) 12 V and 4 A

(D) 4 V and 3 A

13. What is the determinant of this matrix?

$$\begin{bmatrix} 2 & -4 \\ 5 & 1 \end{bmatrix}$$

(A) 22

(B) -18

(C) -40

(D) 20

14. What is the determinant of this matrix?

$$\begin{bmatrix} 1 & 4 & 1 \\ 1 & 2 & 1 \\ 2 & 3 & 1 \end{bmatrix}$$

(A) 2

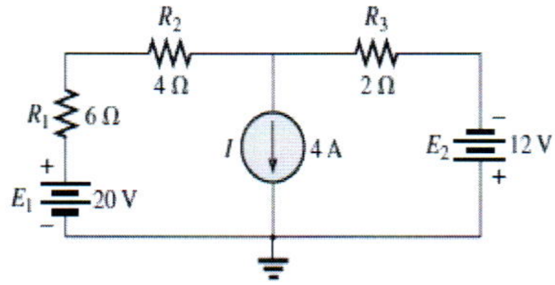
(B) -10

(C) -2

(D) 10

15. What are the values of resistors after converting the delta to star ?

20. Find current I_{R1} of circuit below.



(A) 3.33A (B) 4.0A, (C) 5.77A (D) -2.0A

Part 3 (20 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

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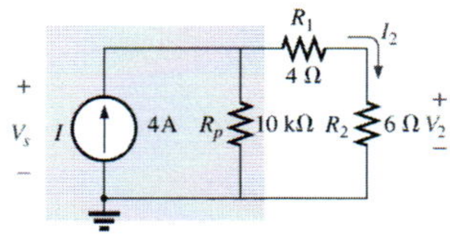
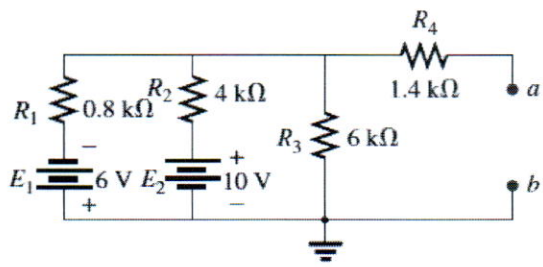
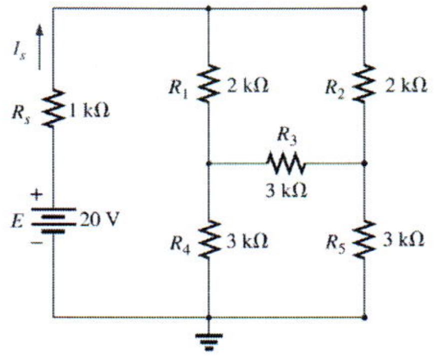
1. (5 points) For the network in below,
   - 1.1 Find current I_2
   - 1.2 Calculate voltage V_2
   - 1.3 Find the source voltage V_S



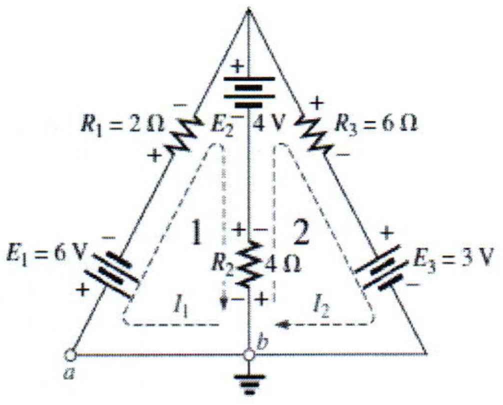
2. (5 points) Find the R_{TH} and E_{th} at terminal *a* and *b* of the figure below.



3. (5 points) Use Δ to Y of 3 kΩ resistors in the network below and find the current I_S.



4. (5 points) Determine the current I_1, and I_2 by using mesh analysis method.



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Answer Sheet

Name:______________________________ ID:________________

Subject: 031021121 Electric Circuits I

Part 1

	True	False
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		

Part 2

	A	B	C	D
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